OrthoStyle: Manifold-Preserving Training-Free Style Transfer via Score-Orthogonal Guidance

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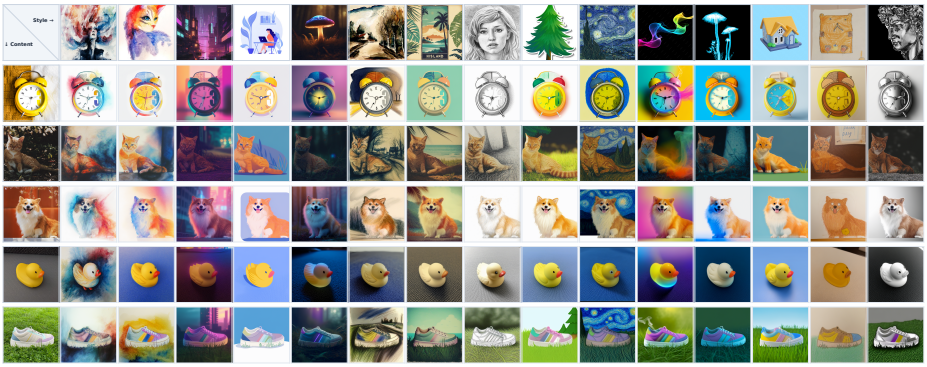

Fig. 1. OrthoStyle across diverse content categories and artistic styles. Rows show diverse content targets, and columns represent artistic styles. Under null-prompt inference, the results preserve subject geometry while capturing rich stroke patterns and textures without semantic leakage.

Abstract. Reference-based diffusion style transfer aims to synthesize artistic textures while preserving target semantic. However, training-free methods face a trade-off: strong style injection causes structural distortion and semantic leakage, whereas conservative transfer yields under-stylization. To address this problem, we introduce **OrthoStyle**, a training-free framework solving this without fine-tuning or DDIM inversion. It combines three components: Semantic Gating to preserve relevant structure; Score-Orthogonal Guidance to project style gradients orthogonally to the diffusion score, minimizing denoising interference; and an Early Step Controller applying AdaIN pushforward to establish global layout before fine stylization. Extensive evaluations demonstrate that OrthoStyle achieves an optimal style-content balance, notably improving style fidelity by 25.9% and reducing perceptual distortion LPIPS by 17.4% over the backbone baseline.

Keywords: Training-Free Style Transfer · Diffusion Models · Manifold-Preserving

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1 Introduction

Text-to-image (T2I) diffusion models [25,23,15] excel at generating high-quality images from text descriptions. As these models are widely adopted in creative applications [13,2], users increasingly demand precise control over both the semantic content and the visual style of the generated images. While semantic content (e.g., "a sitting cat") is easily articulated through text, conveying a specific visual style (e.g., oil painting or watercolor) is inherently abstract and difficult to describe. Text prompts alone are often insufficient to capture these visual nuances due to language constraints. Consequently, reference-based style transfer, which uses a single reference image to guide the visual style has emerged as a crucial task in generative computer vision [14,34,26].

To personalize pre-trained T2I diffusion models for a target artistic style, existing research explores two main directions: optimization-based fine-tuning and training-free modulation. Fine-tuning approaches [27,16,28,29,10] adapt model weights to learn specific styles, but they are computationally expensive and highly prone to semantic overfitting when restricted to a single reference image. Therefore, recent efforts focus on training-free modulation during inference [14,34,6,18,12]. However, training-free methods still face two major challenges: preserving the target structure and transferring strong style without **content leakage**, where semantic elements from the style reference appear in the output. Strong style injection can distort the target object, as in InstantStyle [34] and DiffuseIT [20], while more conservative methods such as ITOC [4] better preserve structure but often produce weak stylization.

To address this trade-off, we present **OrthoStyle**, a training-free style transfer framework built on the Stable Cascade [23]. OrthoStyle combines three complementary mechanisms: **Semantic Gating**, **Score-Orthogonal DINO Guidance**, and an **Early-Step Controller with Blended Style Tokens**. By jointly improving stylization and mitigating structural distortion and semantic leakage, our main contributions are:

- We introduce **OrthoStyle**, a training-free reference-based style transfer framework built on Stable Cascade, requiring neither network fine-tuning nor DDIM inversion.
- **Semantic Gating** refines ControlNet Canny features by filtering background and noisy edges, preserving relevant content structure while allowing greater flexibility for stylization.
- **Score-Orthogonal DINO Guidance** projects DINO-based style gradients orthogonally to the diffusion score direction, reducing interference with the sampling trajectory and structural distortion.
- An **Early-Step Controller with Blended Style Tokens** combines clean-latent AdaIN pushforward with soft style-token blending to establish global appearance while retaining fine-grained stylistic details.
- We also provide theoretical justification for the score-orthogonal projection and conduct extensive quantitative and qualitative evaluations on diverse content and style images, demonstrating a favorable balance between style

fidelity and content preservation compared with both stylization and OC-based methods.

2 Related Work

2.1 Diffusion-Based Style Personalization

Text-to-image (T2I) diffusion models [15,25,23,32] have been widely applied to image editing [13,2], subject-driven generation [27,16], and artistic stylization [14,34,26]. Reference-based personalization methods can be broadly divided into optimization-based approaches, which adapt model parameters to a target concept or style [27,16,28,29,10], and training-free approaches, which modify intermediate features or the sampling process while keeping the pre-trained model fixed [14,34,26,4]. Optimization-based methods require additional adaptation for each target reference, whereas training-free methods avoid retraining but still face a trade-off between style fidelity and content preservation. Motivated by this setting, **OrthoStyle** performs reference-based stylization without network fine-tuning or DDIM inversion.

2.2 Training-Free Reference-Based Style Transfer

Training-free style transfer commonly incorporates reference information through attention features or intermediate representations. Inspired by classical neural style transfer [11,19], StyleAligned [14] shares self-attention features between reference and generated samples and applies AdaIN-based normalization. However, it relies on DDIM inversion [31] to extract reference features, introducing additional inference overhead. InstantStyle [34] avoids explicit inversion by injecting reference features through selected cross-attention layers of IP-Adapter [36]. Other methods, including StyleID [6], DiffStyler [18], and FreeStyle [12], further explore feature- and attention-based modulation for training-free stylization. A key challenge is **content leakage**, where semantic information from the style reference is transferred together with its appearance. Reducing this effect often requires weaker style injection or stronger structural constraints, which may limit stylistic variation. Among OC-based approaches, ITOC [4] performs training-free stylization by optimizing the diffusion trajectory with a style objective while maintaining fidelity to the source image. Recently, TRACE [21] introduces time-adaptive guidance, explored how to disentangle content and style, and proposes Directional Content Leakage (DCL) to quantify semantic leakage. RB-Modulation [26] formulates style transfer on Stable Cascade [23] as an inference-time control problem and spatially aggregates style tokens to reduce semantic transfer. However, aggregation can suppress token variation and weaken fine-grained texture representation. In addition, raw Canny conditioning may retain irrelevant background edges and impose unnecessary structural constraints. Based on these observations, we design two complementary components: **Semantic Gating** to retain relevant structural cues, and **Blended Style Tokens** to preserve reference variation while reducing semantic interference from raw style tokens.

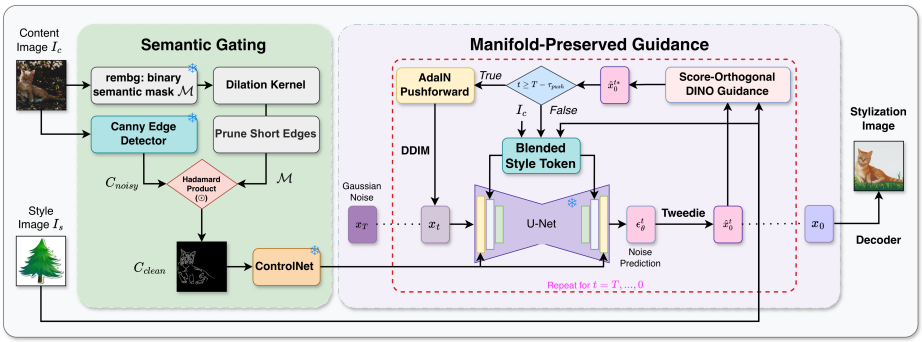

Fig. 2. Overview of the OrthoStyle pipeline. Given a content image I_c and a style reference I_s, OrthoStyle combines three components: (1) **Semantic Gating** to refine structural cues; (2) an **Early-Step Controller with Blended Style Tokens** to establish global appearance and inject style information across denoising stages; and (3) **Score-Orthogonal DINO Guidance** to improve style similarity while reducing interference with the diffusion score direction.

2.3 Guidance and Timestep-Aware Diffusion Control

Guidance methods steer the reverse diffusion process toward a desired objective. Classifier guidance [7] introduces gradients from an auxiliary model, while classifier-free guidance combines conditional and unconditional predictions. More general inference-time methods, such as Universal Guidance [1] and Diffusion Self-Guidance [9], further show that external objectives or internal diffusion representations can control generation without retraining. Tweedie's formula [8] estimates the clean sample $\mathbf{x}_0$ from noisy states, enabling objective evaluation in the predicted clean space. Related methods such as DPS [5] and TFG [35] similarly apply gradient-based corrections during reverse diffusion. However, an external guidance gradient is not necessarily aligned with the diffusion score, and directly applying it may interfere with the denoising trajectory. This motivates our **Score-Orthogonal DINO Guidance**, which removes the component of the style gradient parallel to the diffusion score before applying the update. Diffusion control is also timestep-dependent. SDEdit [22], Prompt-to-Prompt [13], and Plug-and-Play Diffusion Features [33] show that interventions at different denoising stages can affect structure and appearance differently. Consequently, our **Early-Step Controller** applies clean-latent AdaIN pushforward during initial steps to establish global appearance before fine-grained stylization.

3 Methodology

The overall pipeline of **OrthoStyle** is illustrated in Fig. 2. Given a content image I_c and a style reference I_s, OrthoStyle generates a stylized output in a fully training-free manner.

3.1 Semantic Gating

Structural conditioning is important for preserving the geometry of I_c. However, a raw Canny map retains all detected edges, including irrelevant background structures, which may over-constrain the generation process and limit stylization. We therefore introduce **Semantic Gating** to retain structural cues associated with the foreground subject. Specifically, we extract a foreground mask $\mathcal{M} = \text{rembg}(I_c)$ and use its dilated region to filter the raw Canny map C_{noisy} as defined in Eq. (1):

$$C_{\text{clean}} = C_{\text{noisy}} \odot \text{Dilation}(\mathcal{M}, \text{kernel} = 5). \tag{1}$$

We remove the smallest 20% of detected edges to reduce structural constraints. The resulting map C_{clean} conditions ControlNet with strength λ_{cnet}.

3.2 Score-Orthogonal DINO Guidance

To mitigate the semantic interference and structural distortion caused by aggressive token injection, we introduce an auxiliary perceptual objective to guide the predicted clean sample $\hat{x}_{0|t}$ toward the reference style. Given the diffusion score $\mathbf{s}_t = \mathbf{s}_\theta(\mathbf{x}_t, t; \mathbf{c}_t, C_{\text{clean}})$, we employ a frozen DINO-ViT-S/16 [3] encoder Ψ_{DINO} to define the style loss in Eq. (2):

$$\mathcal{L}_{\text{style}} = \left\| \Psi_{\text{DINO}}(\hat{x}_{0|t}) - \Psi_{\text{DINO}}(I_s) \right\|_2^2. \tag{2}$$

As diffusion in Stable Cascade operates in latent space $\mathbf{z}_t$, latent representations cannot directly evaluate perceptual style metrics, unlike the RGB-trained DINO encoder Ψ_{DINO}. We thus decode clean latent $\hat{\mathbf{z}}_{0|t}$ to pixel space $\hat{x}_{0|t} = \mathcal{D}(\hat{\mathbf{z}}_{0|t})$ via the pre-trained VAE decoder $\mathcal{D}$, propagating style gradients via $\nabla_{\hat{\mathbf{z}}_{0|t}} \mathcal{L}_{\text{style}} = \mathcal{J}_{\mathcal{D}}(\hat{\mathbf{z}}_{0|t})^\top \nabla_{\hat{x}_{0|t}} \mathcal{L}_{\text{style}}$ (with decoder Jacobian $\mathcal{J}_{\mathcal{D}}$). The resulting style-guidance direction is formulated as Eq. (3):

$$u_t = -\eta \nabla_{\hat{x}_{0|t}} \mathcal{L}_{\text{style}}, \tag{3}$$

where η controls the guidance strength. Directly applying u_t may interfere with the diffusion score direction. We therefore remove its score-aligned component using Eq. (4):

$$u_t^\perp = u_t - \frac{\langle u_t, \mathbf{s}_t \rangle}{\|\mathbf{s}_t\|_2^2} \mathbf{s}_t, \tag{4}$$

and update the predicted clean sample as shown in Eq. (5):

$$\hat{x}_{0|t}^* \leftarrow \hat{x}_{0|t}^* + u_t^\perp. \tag{5}$$

In practice, a small ϵ_{safe} is added to the denominator of Eq. (4) for numerical stability.

Theoretical Justification. The projection in Eq. (4) provides two useful first-order properties. First, by construction,

$$\langle u_t^{\perp}, \mathbf{s}_t \rangle = 0. \tag{6}$$

Since the score represents the local direction of increasing log-density, an orthogonal update lies along its local level set to first order. For a small update magnitude δ, a Taylor expansion gives

$$\begin{aligned} \log p_t(\mathbf{x}_t + \delta u_t^{\perp}) - \log p_t(\mathbf{x}_t) = \delta \langle \mathbf{s}_t, u_t^{\perp} \rangle \\ + \frac{\delta^2}{2} (u_t^{\perp})^{\top} \mathbf{H}_t u_t^{\perp} + \mathcal{O}(\delta^3), \end{aligned} \tag{7}$$

where $\mathbf{H}_t = \nabla^2_{\mathbf{x}_t} \log p_t(\mathbf{x}_t)$. Because the first-order term vanishes, the change in local log-density is governed only by second-order terms. This provides a local geometric interpretation of score-orthogonal guidance as reducing first-order deviation from the diffusion trajectory.

Second, the projected direction remains a descent direction for the style objective. Since $u_t = -\eta \nabla_{\hat{x}_{0|t}} \mathcal{L}_{\text{style}}$, we have

$$\left\langle \nabla_{\hat{x}_{0|t}} \mathcal{L}_{\text{style}}, u_t^{\perp} \right\rangle = -\frac{1}{\eta} \|u_t^{\perp}\|_2^2 \leq 0. \tag{8}$$

Therefore, the projection removes the component aligned with the diffusion score while retaining a first-order descent direction for style optimization. This provides the geometric motivation for improving style similarity with reduced interference with the underlying denoising process.

3.3 Early-Step Controller with Blended Style Tokens

Full token injection risks semantic leakage, whereas complete averaging destroys spatial details vital for fine stylization. To resolve this trade-off, we introduce **Blended Style Tokens**.

Let $\mathbf{c}_{\text{style}} \in \mathbb{R}^{L_{\text{clip}} \times C}$ denote the style-token sequence extracted from I_s. We construct the blended representation according to Eq. (9):

$$\mathbf{c}_{\text{blended}} = \alpha_s \mathbf{c}_{\text{style}} + (1 - \alpha_s) \left(\frac{1}{L_{\text{clip}}} \sum_{j=1}^{L_{\text{clip}}} \mathbf{c}_{\text{style}}^{(j)} \right), \tag{9}$$

where α_s is a balance ratio. The first term preserves token-level style variation, while the mean component reduces local semantic variation from the reference image.

Style control is also timestep-dependent. Since the early denoising steps have a stronger influence on global appearance, we introduce an **Early-Step Controller** that applies AdaIN [19] to the predicted clean sample during the initial

Algorithm 1 OrthoStyle Pseudocode

Require: Content I_c, style I_s, sampling steps T, score network $\mathbf{s}_\theta$, style descriptor Ψ_{DINO}, balance ratio α_s, pushforward threshold τ_{push}, step size η

Ensure: Stylization image I_{out}

$C_{\text{clean}} \leftarrow \text{SemanticGating}(I_c)$
$\mathbf{c}_{\text{style}} \leftarrow \text{EncodeStyle}(I_s)$
$\mathbf{c}_{\text{base}} \leftarrow [\mathbf{c}_{\text{content}}, \mathbf{c}_{\text{content-style}}, \mathbf{c}_{\text{txt}}]$
$\mathbf{c}_{\text{blended}} \leftarrow \alpha_s \mathbf{c}_{\text{style}} + (1-\alpha_s)\text{Mean}(\mathbf{c}_{\text{style}})$
$\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
for $t = T$ **down to** 1 **do**
  $\mathbf{c}_t \leftarrow \begin{cases} \mathbf{c}_{\text{base}}, & t \geq T - \tau_{\text{push}}, \\ [\mathbf{c}_{\text{base}}, \mathbf{c}_{\text{blended}}], & \text{otherwise} \end{cases}$
  $\mathbf{s}_t \leftarrow \mathbf{s}_\theta(\mathbf{x}_t, t; \mathbf{c}_t, C_{\text{clean}})$
  $\hat{x}_{0|t} \leftarrow \text{PredictClean}(\mathbf{x}_t, \mathbf{s}_t, t)$
  $\mathcal{L}_{\text{style}} \leftarrow \|\Psi_{\text{DINO}}(\hat{x}_{0|t}) - \Psi_{\text{DINO}}(I_s)\|_2^2$
  $u_t \leftarrow -\eta \nabla_{\hat{x}_{0|t}} \mathcal{L}_{\text{style}}$
  $u_t^\perp \leftarrow \text{Orthogonalize}(u_t, \mathbf{s}_t)$
  $\hat{x}^*_{0|t} \leftarrow \hat{x}_{0|t} + u_t^\perp$
  if $t \geq T - \tau_{\text{push}}$ **then**
    $\hat{x}^*_{0|t} \leftarrow \text{AdaINPush}(\hat{x}^*_{0|t}, I_s)$
  end if
  $\mathbf{x}_{t-1} \leftarrow \text{DDIM}(\mathbf{x}_t, \hat{x}^*_{0|t}, \mathbf{c}_t, C_{\text{clean}})$
end for
return I_{out}

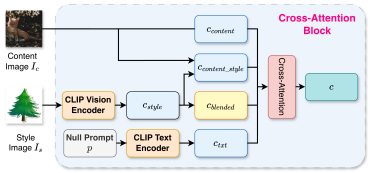

Fig. 3. Blended Style Tokens. The style reference I_s is encoded into $\mathbf{c}_{\text{style}}$ and softly blended with its mean representation to obtain $\mathbf{c}_{\text{blended}}$. This preserves token variation while reducing semantic information that may cause content leakage. The blended representation is concatenated with the base conditioning $\mathbf{c}_{\text{base}}$ and injected through cross-attention. As shown in Alg. 1, early steps use only $\mathbf{c}_{\text{base}}$, while $\mathbf{c}_{\text{blended}}$ is introduced later for fine-grained stylization.

τ_{push} steps. Given the current clean prediction $\hat{x}^*_{0|t}$ (obtained after the score-orthogonal DINO guidance step described in Sec. 3.2) and the style reference I_s, we perform the pushforward operation defined in Eq. (10):

$$\hat{x}^*_{0|t} \leftarrow (1-\gamma_{\text{push}})\hat{x}^*_{0|t} + \gamma_{\text{push}} \left[\frac{\hat{x}^*_{0|t} - \boldsymbol{\mu}(\hat{x}^*_{0|t})}{\boldsymbol{\sigma}(\hat{x}^*_{0|t}) + \epsilon} \odot \boldsymbol{\sigma}(I_s) + \boldsymbol{\mu}(I_s) \right]. \tag{10}$$

Here, $\boldsymbol{\mu}(\cdot)$ and $\boldsymbol{\sigma}(\cdot)$ denote the spatial mean and standard deviation of the features, and ϵ is a small constant to prevent division by zero. With $\gamma_{\text{push}} = 0.60$. The transformation aligns the global channel statistics of the clean prediction with those of the style reference during the early denoising stage, while later steps refine finer stylistic details.

4 Experimental Setup

This section presents the experimental setup, including implementation configurations, benchmark datasets, evaluation metrics, and the baseline methods used for comparison.

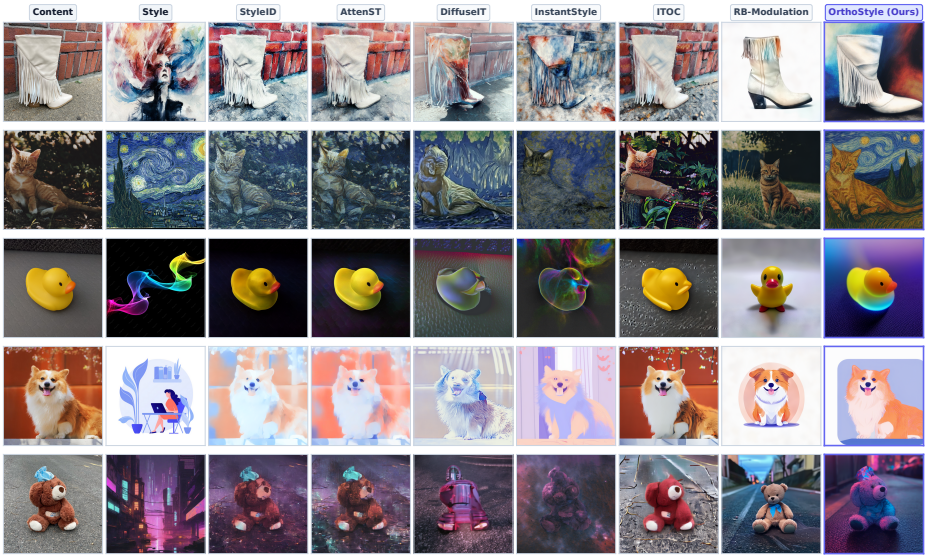

Fig. 4. We compare **OrthoStyle** with representative stylization methods (StyleID, AttenST, DiffuseIT, and InstantStyle) and optimal-control-based methods (ITOC and RB-Modulation) under null-prompt inference. OrthoStyle achieves a better balance between content preservation and stylistic transfer, with fewer structural distortions and reduced semantic leakage.

Configurations: We build OrthoStyle on the Stable Cascade architecture [23]. All experiments are conducted on a single NVIDIA RTX 4090 GPU (24GB VRAM). Across all experiments, the image is sampled for 16 steps with a cosine log-SNR schedule (CFG = 4.0, shift = 2.0). The ControlNet strength is set to $\lambda_{\text{cnet}} = 0.8$. We use DINO-ViT-S/16 for style guidance with a guidance strength $\eta = 0.20$. The Early-Step Controller uses $\gamma_{\text{push}} = 0.60$ for $\tau_{\text{push}} = 1$ step, and the style-token blending factor is set to $\alpha_s = 0.85$.

Benchmark Dataset: Following the standard evaluation protocol established in recent diffusion-based stylization literature [6,26,14], we evaluate OrthoStyle on 225 content-style pairs formed by 15 content images and 15 style references. The content images are selected from the DreamBooth dataset [27], covering animals, vehicles, architecture, and common objects. The style references are drawn from the evaluation sets of RB-Modulation [26] and StyleDrop [29], including diverse artistic appearances such as oil painting, mosaic, woodblock print, and cyberpunk styles. We consider three prompting settings: *Null prompt* (`""`), *Object prompt* (`"a <object>"`), and *Style prompt* (`"a <object> in <style> style"`) for ablation studies.

Evaluation Metrics: We evaluate OrthoStyle using five metrics across three dimensions: **style fidelity**, measured by CSD [30] and DINO-Style [3]; **content preservation**, measured by CLIP-I [24] and LPIPS [37]; and **content leak-**

Table 1. Quantitative comparison on the 225-pair benchmark. Best and second-best results are shown in **bold** and underlined, respectively.

Method	Style Fidelity		Content Preservation		Leakage
	CSD ↑	DINO ↑	CLIP-I ↑	LPIPS ↓	DCL ↓
Stylization Methods					
OrthoStyle	39.49	23.86	**79.87**	56.22	26.41
StyleID [6]	45.11	26.95	77.32	45.97	27.05
AttenST [17]	41.44	24.80	77.17	**43.47**	**26.07**
DiffuseIT [20]	31.90	**32.11**	68.28	54.43	42.69
InstantStyle [34]	**47.67**	27.80	68.27	61.55	36.49
OC-Based Methods					
OrthoStyle	**39.49**	**23.86**	79.87	56.22	26.41
ITOC [4]	14.62	16.04	**86.57**	**33.86**	**19.39**
RB-Modulation [26]	31.38	19.13	81.69	68.09	28.81

age, measured by DCL [21]. Higher values are better for CSD, DINO-Style, and CLIP-I, while lower values are preferred for LPIPS and DCL.

Comparison Methods: We compare OrthoStyle with two groups of training-free baselines: **stylization methods**, including StyleID [6], AttenST [17], DiffuseIT [20], and InstantStyle [34]; and **OC-based methods**, including ITOC [4] and RB-Modulation [26]. All methods are evaluated on the same 225 content-style pairs under identical null-prompt conditions.

5 Results

5.1 Qualitative Results

Fig. 4 compares **OrthoStyle** with representative training-free baselines. OC-based methods generally preserve content structure but produce weaker stylization, whereas stylization-oriented methods achieve stronger style transfer at the cost of structural distortion or semantic leakage. In contrast, our method better balances these objectives by preserving the target geometry while remaining closer to the reference style. Semantic Gating protects relevant structure, while the Early-Step Controller with Blended Style Tokens and Score-Orthogonal DINO Guidance introduce style information with reduced structural distortion and semantic leakage.

5.2 Quantitative Results

Based on Tab. 1, we analyze the performance of **OrthoStyle** against two primary categories of baselines:

Comparison with Stylization Methods: For stylization methods, strong style fidelity often sacrifices content preservation. InstantStyle achieves the highest CSD (47.67), while DiffuseIT obtains the highest DINO score (32.11); however, both exhibit considerably lower CLIP-I (68.27 and 68.28) and higher DCL (36.49 and 42.69). By comparison, **OrthoStyle** achieves the highest CLIP-I (79.87) and the second-lowest DCL (26.41) in this group, while retaining competitive style fidelity. This confirms our mechanisms effectively balance style transfer and content preservation.

Comparison with OC-Based Methods. OC-based methods favor content preservation over stylization. ITOC and RB-Modulation achieve higher CLIP-I scores (86.57 and 81.69), but substantially lower style fidelity, particularly ITOC (CSD 14.62, DINO 16.04). Compared with RB-Modulation, which uses the same Stable Cascade backbone, **OrthoStyle** improves CSD from 31.38 to 39.49 and DINO from 19.13 to 23.86. At the same time, DCL decreases from 28.81 to 26.41, indicating reduced leakage despite stronger stylization. Overall, our method enhances style transfer over the closest OC baseline while maintaining comparable content preservation.

Pareto Trade-off: Although **OrthoStyle** does not achieve the best score on every metric, Fig. 5 shows a favorable balance between style fidelity and content preservation. Methods with stronger style scores often show lower content similarity or higher leakage, whereas content-focused methods tend to produce weaker stylization. By constraining style guidance through score-orthogonal projection, our method improves style alignment while limiting interference with denoising. This places OrthoStyle in a competitive region of the style-content trade-off rather than favoring one objective over the other.

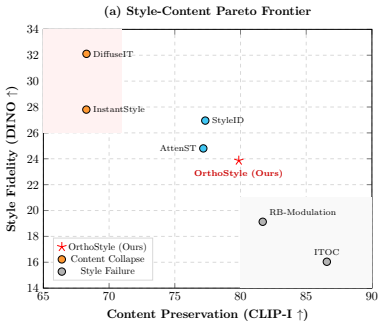

Fig. 5. Style-Content Pareto Frontier. OrthoStyle occupies the optimal trade-off, avoiding both content collapse and under-stylization.

6 Ablation Study

To assess the contribution of each component, we conduct ablation studies on all 225 content-style pairs. Quantitative results are reported in Tab. 2, while qualitative comparisons are shown in Fig. 6.

6.1 Impact of Blended Style Tokens (α_s)

This ablation examines whether blending style tokens helps balance style fidelity and content preservation. With pure mean pooling ($\alpha_s = 0.0$, Configuration B), style information is strongly suppressed, reducing CSD from 39.49 to 23.35

Table 2. The table evaluates the contribution of each component, the effect of the pushforward threshold τ, and different prompt-conditioning levels. All experiments are evaluated on 225 content-style pairs, except the pushforward sweep, which uses a 7×7 subset. The default setting is $\alpha_s = 0.85$ and $\tau = 1$ unless otherwise specified.

Configuration	CSD ↑	DINO ↑	CLIP-I ↑	LPIPS ↓	DCL ↓
OrthoStyle	39.49	23.86	79.87	56.22	26.41
(B) Pure Mean Token ($\alpha_s = 0.00$)	23.35	17.56	85.15	55.04	20.58
(C) Raw Style Token ($\alpha_s = 1.00$)	52.64	34.38	68.94	67.92	44.82
(D) w/o Score-Orthogonal Guidance	46.66	30.96	70.80	70.19	43.26
(E) w/o AdaIN Pushforward	65.87	49.53	58.74	74.89	64.74
(F) w/o Semantic Gated Canny	44.98	30.36	71.13	69.88	42.23
Prompt Conditioning Levels					
Prompt: Object Level	29.10	18.92	83.82	55.23	21.57
Prompt: Style Level	40.23	25.21	78.01	59.98	30.00
Pushforward Step Threshold					
$\tau = 1$	45.14	27.34	75.90	58.55	27.27
$\tau = 2$	41.57	25.51	76.50	56.61	25.50
$\tau = 3$	40.40	24.54	78.44	55.51	23.52
$\tau = 4$	37.07	23.27	79.40	54.22	21.65

and DINO from 23.86 to 17.56. In contrast, using raw style tokens ($\alpha_s = 1.0$, Configuration C) increases style fidelity substantially, but CLIP-I drops to 68.94 and DCL rises to 44.82, indicating stronger content distortion and leakage from the style reference. The blended setting ($\alpha_s = 0.85$) provides a better balance, preserving stylistic variation while suppressing semantic transfer.

6.2 Effect of Score-Orthogonal Guidance

This ablation evaluates whether score-orthogonal projection helps introduce style guidance without distorting content. Removing the projection (Configuration D) increases CSD and DINO to 46.66 and 30.96, but substantially reduces CLIP-I from 79.87 to 70.80 and increases DCL from 26.41 to 43.26. While unconstrained style gradients improve stylization at the expense of structure and leakage, Score-Orthogonal Guidance mitigates this interference by projecting out the score-aligned component, optimally balancing style and content.

6.3 Role of Clean-Latent AdaIN Pushforward

This experiment studies whether the early AdaIN pushforward is important for establishing a stable global appearance before fine-grained stylization. Without AdaIN Pushforward (Configuration E), style scores increase considerably, with CSD reaching 65.87 and DINO reaching 49.53, but content preservation deteriorates sharply: CLIP-I decreases to 58.74 and DCL increases to 64.74. These results indicate that strong style injection without early global control can easily

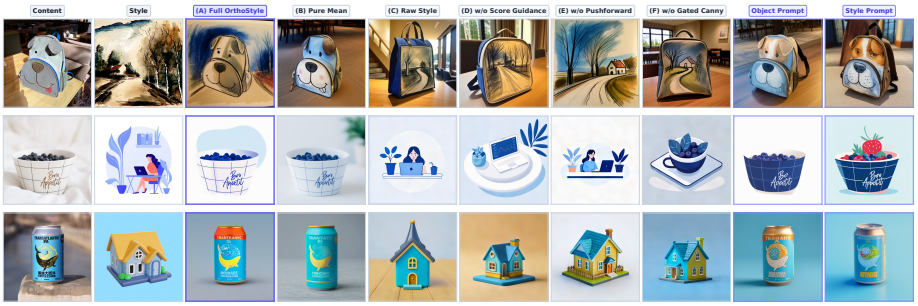

Fig. 6. Qualitative ablation study. Columns (A)-(F) compare the full OrthoStyle and individual component variants under the null prompt (""). The last two columns show Object-level prompt ("a <object>") and Style-level prompt ("a <object> in <style> style") conditioning.

dominate the original content. Applying AdaIN to the predicted clean sample during the early steps helps establish the global appearance of the style reference while retaining more of the target structure.

6.4 Effect of Semantic Gating

This ablation verifies whether filtering irrelevant Canny edges improves structural conditioning. Without Semantic Gating (Configuration F), CSD and DINO increase to 44.98 and 30.36, but CLIP-I decreases from 79.87 to 71.13 and DCL increases from 26.41 to 42.23. This suggests that using all detected edges does not necessarily improve content preservation, since background contours and irrelevant structures can introduce unnecessary constraints during generation. Semantic Gating focuses the structural condition on relevant foreground regions, resulting in better content preservation and lower leakage.

6.5 Controllability via the Pushforward Threshold τ

This experiment examines whether the pushforward threshold τ can control the trade-off between stylization and content preservation at inference time. As τ increases from 1 to 4, CLIP-I consistently improves from 75.90 to 79.40, while DCL decreases from 27.27 to 21.65. At the same time, style fidelity gradually decreases, with CSD changing from 45.14 to 37.07 and DINO from 27.34 to 23.27. Therefore, τ provides a simple inference-time control: smaller values favor stronger stylization, whereas larger values place greater emphasis on content preservation, without requiring model retraining.

6.6 Effect of Prompt Conditioning

This experiment studies how prompt conditioning affects the style-content trade-off. Object-level prompts improve content preservation (CLIP-I 83.82, DCL 21.57) but weaken stylization, while Style-level prompts increase style fidelity (CSD 40.23, DINO 25.21) at the cost of higher leakage. The null prompt provides a more balanced trade-off between these objectives.

7 Conclusion

In this work, we introduce **OrthoStyle**, a training-free framework for reference-based diffusion style transfer that jointly improves style fidelity and content preservation while mitigating semantic leakage. To achieve this balance, **Semantic Gating** first refines structural conditioning by suppressing irrelevant background cues while preserving the underlying geometry. Building on this cleaner structural prior, **Score-Orthogonal DINO Guidance** introduces perceptual style guidance while projecting out its score-aligned component, thereby reducing interference with the original denoising trajectory. Complementing this guidance, an **Early-Step Controller with Blended Style Tokens** establishes the global stylistic appearance during early denoising steps while preserving fine-grained content details at later stages. Together, these components enable **OrthoStyle** to effectively balance style fidelity, content preservation, and semantic leakage, as supported by our theoretical analysis and evaluations on 225 image pairs against existing baselines.

Limitations and Future Work. A current limitation of Semantic Gating is its reliance on salient foreground extraction. When processing complex scenes without an unambiguous foreground subject (e.g., wide panoramic landscapes), the segmentation mask may saturate. In future work, we plan to integrate prompt-guided foundation segmentation models such as Segment Anything (SAM) to achieve adaptive and precise structural gating across arbitrary scenes.

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